ALGEBRA 2 · SEQUENCES & SERIES

Sequence Builder

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Sequence Builder** on Mr. K's Classroom: Algebra 2 → Sequences & Series → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Starting at 1, which grows faster: adding 5 each time or multiplying by 2 each time?

   Multiplying by 2 (after a few terms).

2 Explore use the tool and record what you see

Identify each sequence

Rebuild each sequence in the tool.

Sequence	Type	d or r	Next term	Formula
3, 7, 11, 15, …	**arithmetic**	**d = 4**	**19**	$a_n = 3 + 4(n - 1)$
5, 10, 20, 40, …	**geometric**	**r = 2**	**80**	$a_n = 5 \cdot 2^{n-1}$
20, 17, 14, 11, …	**arithmetic**	**d = −3**	**8**	$a_n = 20 - 3(n - 1)$
8, −4, 2, −1, …	**geometric**	$r = -\frac{1}{2}$	$\frac{1}{2}$	$a_n = 8(-\frac{1}{2})^{n-1}$
16, 8, 4, 2, …	**geometric**	$r = \frac{1}{2}$	**1**	$a_n = 16(\frac{1}{2})^{n-1}$

Running sums for $a_1 = 8$, $r = \frac{1}{2}$

n	1	2	3	4	5	→ ∞
S_n	8	**12**	**14**	**15**	**15.5**	**16**

3 Write the rule fill in each blank from what you observed

1. Arithmetic sequences **add** the same number each time; their terms lie on a **line**.
2. Geometric sequences **multiply** by the same number each time; their terms follow an **exponential** curve.
3. A negative common ratio makes the terms **alternate** signs.
4. An infinite geometric series has a sum only when **$|r| < 1$**, and then the sum is $\frac{a_1}{(1 - r)}$.

4 Check no tool: show your work

1. Find the 10th term of 4, 9, 14, ...

   49

2. Find the 6th term of 3, 6, 12, ...

   96

3. Find the sum of 10 + 5 + 2.5 + ...

   20

4. Does 2 + 6 + 18 + ... have a sum?

   No: r = 3, so it grows without bound

5 Reflect

Why does a geometric series with r = $\frac{1}{2}$ stop growing past a certain value?

Each new term is half the one before, so the amounts added keep shrinking and the total closes in on a limit.